Symbol	Unit	Meaning
Indices		
$i,\ o$	Token	Input (sold) and output (bought) endpoint tokens, respectively; (i,o) is an ordered input–output pair and the direct route is $i \to o$.
k	Token	Candidate vehicle token used as the route intermediate; $k \notin \{i,o\}$.
h	Token	Candidate vehicle token used as a challenger to incumbent k; $h \in \mathcal{K} \setminus \{i,o,k\}$.
$\ell,\ p,\ p'$	Token / pool	ℓ indexes tokens in cross-token sums; p indexes a focal liquidity pool; p' indexes another pool in leave-one-out sums.
$t,\ u,\ w$	UTC day / UTC week	t indexes a focal calendar day; u indexes another calendar day in a time-window sum; w indexes calendar weeks in settlement variables.
τ	Days	Positive calendar-day horizon selected ex ante for each dynamic specification.
$d,\ \mu$	Calendar days / calendar months	d is calendar-day event time relative to a candidate-specific shock; μ is calendar-month event time relative to an architecture activation date. Event time zero contains the event or activation date.
g	Settlement comparison cell	Settlement comparison cell defined by ordered pair (i,o), vehicle k, UTC week w, and a prespecified route-size bin.
q	USD	Input quote notional. Un-suffixed route-cost columns use $q = \$10{,}000$.
r	Route unit	Reconstructed input-to-output route unit. A coherent $i \to k \to o$ component contributes one r regardless of its number of legs; a split or join contributes one r per reconstructed input–output pair.
Route and liquidity aggregates		
$\mathrm{Vol}_t,\ \mathrm{DVol}_t$	USD	Vol_t is total realized USD volume across all direct and indirect route units on day t. DVol_t restricts it to direct routes, so $0 \le \mathrm{DVol}_t \le \mathrm{Vol}_t$.
$\mathrm{IVol}_t,\ \mathrm{IVol}_{k,t}$	USD	IVol_t restricts Vol_t to indirect routes, so $0 \le \mathrm{IVol}_t \le \mathrm{Vol}_t$. $\mathrm{IVol}_{k,t}$ further restricts that volume to routes using vehicle k, so $0 \le \mathrm{IVol}_{k,t} \le \mathrm{IVol}_t$.
$\mathrm{Vol}_{i,o,t},\ \mathrm{IVol}_{i,o,t},\ \mathrm{IVol}_{i,o,k,t}$	USD	$\mathrm{Vol}_{i,o,t}$ is realized route-unit volume for ordered pair (i,o) on day t. $\mathrm{IVol}_{i,o,t}$ restricts it to indirect routes, and $\mathrm{IVol}_{i,o,k,t}$ further restricts it to routes through k, so $0 \le \mathrm{IVol}_{i,o,k,t} \le \mathrm{IVol}_{i,o,t} \le \mathrm{Vol}_{i,o,t}$.
$N_t,\ N^{\mathrm{in}}_{k,t},\ N^{\mathrm{out}}_{k,t}$	Route-unit count	N_t counts all reconstructed route units r on day t, not their individual legs. $N^{\mathrm{in}}_{k,t}$ and $N^{\mathrm{out}}_{k,t}$ restrict that count to routes with k as the input or output endpoint, respectively; each lies between 0 and N_t.
$N^I_t,\ N^I_{k,t}$	Route-unit count	N^I_t restricts N_t to route units with at least one intermediate token. $N^I_{k,t}$ further restricts that count to units with intermediate k, so $0 \le N^I_{k,t} \le N^I_t \le N_t$; superscript I denotes indirect routes.
$\mathcal{A}_t,\ \mathcal{A}^k_t,\ \mathcal{M}^k_t$	Sets of token pairs	$\mathcal{A}_t$ is the set of endpoint pairs active in indirect routes on day t. $\mathcal{A}^k_t$ restricts it to pairs using k, and $\mathcal{M}^k_t$ further restricts it to pairs for which k has the largest candidate-vehicle volume; $\mathcal{M}^k_t \subseteq \mathcal{A}^k_t \subseteq \mathcal{A}_t$.
$\mathrm{LegVol}^{\mathrm{in}}_{k,t},\ \mathrm{LegVol}^{\mathrm{out}}_{k,t}$	USD	$\mathrm{LegVol}^{\mathrm{in}}_{k,t}$ and $\mathrm{LegVol}^{\mathrm{out}}_{k,t}$ are route-leg USD volumes for which k is the leg's input or output token, respectively, on day t.
$\mathrm{Vol}^{\mathrm{in}}_{k,t},\ \mathrm{Vol}^{\mathrm{out}}_{k,t}$	USD	Input- and output-endpoint restrictions of Vol_t for token k: $0 \le \mathrm{Vol}^{\mathrm{in}}_{k,t} \le \mathrm{Vol}_t$ and $0 \le \mathrm{Vol}^{\mathrm{out}}_{k,t} \le \mathrm{Vol}_t$.
$\mathcal{K}$	Set of tokens	Prespecified candidate set $\{\mathrm{WETH}, \mathrm{USDC}, \mathrm{USDT}, \mathrm{DAI}, \mathrm{WBTC}\}$.
$\mathcal{L}_t,\ \mathcal{L}_{k,t},\ m_p$	Set of pools / token count	$\mathcal{L}_t$ contains valid Uniswap V3 daily-snapshot pools with exact token contracts identified in the persisted V3 swap archive and $0 < \mathrm{TVL}_{p,t} \le 10$ billion USD. $\mathcal{L}_{k,t} \subseteq \mathcal{L}_t$ restricts that set to pools with candidate k on one side. m_p is the number of pool tokens in $\mathcal{K}$, so $m_p \in \{1,2\}$.
$\mathrm{TVL}_{p,t},\ L_{k,t}$	USD	$\mathrm{TVL}_{p,t}$ is pool p's day-t USD TVL from the V3 daily snapshot. $L_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} \mathrm{TVL}_{p,t}/m_p$; a one-candidate pool contributes all TVL to that candidate and a two-candidate pool contributes one half to each.
R^{WETH}_t	Daily log-return fraction	Day-t log return of the WETH price used to construct downside stress.
Candidate-shock objects		
$P_{k,t}$	USD per token	Day-t USD price of candidate token k.
$R_{k,t}$	Daily log-return fraction	Candidate return $R_{k,t} = \ln(P_{k,t}/P_{k,t-1})$.
$\sigma^{(30)}_{k,t-1}$	Daily log-return standard deviation	Sample standard deviation of $R_{k,u}$ over the 30 calendar days ending at $t-1$, requiring at least 20 valid daily returns.
Persistence and displacement objects		
$k^\star_{i,o,t},\ h^\star_{i,o,q,t}$	Token	$k^\star_{i,o,t}$ is the incumbent vehicle with the largest mean $\mathrm{VehicleShare}_{i,o,k,u}$ over the 30 calendar days ending at $t-1$. $h^\star_{i,o,q,t}$ is the executable nonincumbent candidate with the largest $O^I_{i,o,h,q,t}$ on day t.
Architecture objects		
$t^{\mathrm{V3}}_0,\ \mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}},\ \mathcal{P}^{\mathrm{V3}}_q$	UTC day / set of UTC days / set of token pairs	t^{V3}_0 is the Uniswap V3 activation date, 5 May 2021. $\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}$ is the fixed 180-calendar-day window ending at $t^{\mathrm{V3}}_0 - 1$. $\mathcal{P}^{\mathrm{V3}}_q$ is the fixed architecture-analysis universe of ordered pairs with positive realized route volume on at least 30 days in that pre-period; every member is quoted at q throughout the event window, independent of post-V3 activity.
Route-cost quote objects		
$\mathcal{P}_{k,t,q}$	Set of token pairs	Quote-universe pairs for candidate k: distinct ordered input–output pairs among day t's 200 largest clean reconstructed pairs by realized USD volume, where clean means route class `single` or `coherent`, $k \notin \{i,o\}$, and each of i, o, and k has a valid day-price estimate. A valid estimate requires at least three finite token-side USD-per-token observations in $(0, \$1{,}000{,}000)$ and equals their realized-USD-volume-weighted median. Each pair is submitted to the direct and via-k quote engines at input q; membership does not require either quote to execute.
$\mathcal{D}_{k,t,q},\ \mathcal{I}_{k,t,q}$	Sets of token pairs	$\mathcal{D}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ and $\mathcal{I}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ restrict the broad pair set to pairs with an executable direct route or an executable indirect route via k.
$\mathcal{C}_{k,t,q}$	Set of token pairs	Common-support pairs for candidate k, day t, and notional q: $\mathcal{C}_{k,t,q} = \mathcal{D}_{k,t,q} \cap \mathcal{I}_{k,t,q}$, so both the direct and indirect routes are executable.
$\mathcal{T}_{k,t,q},\ \mathcal{W}_{k,t,q}$	Sets of token pairs	$\mathcal{T}_{k,t,q} \subseteq \mathcal{D}_{k,t,q}$ is the thin-direct subset. $\mathcal{W}_{k,t,q} \subseteq \mathcal{C}_{k,t,q}$ contains common-support pairs for which the indirect route beats the direct route, equivalently $\Delta C^D_{i,o,k,q,t} < 0$.
$D_{i,o,q,t},\ I_{i,o,k,q,t},\ T_{i,o,q,t}$	Indicator (0/1)	Indicators for direct-route availability (D), indirect-route-through-k availability (I), and an executable direct route returning less than $0.9q$ (T).
$O^D_{i,o,q,t},\ O^I_{i,o,k,q,t}$	USD	Quoted output values; superscripts D and I denote direct and indirect routes.
$\Delta C^D_{i,o,k,q,t}$	Fraction	Pair-level direct cost advantage $(O^D_{i,o,q,t} - O^I_{i,o,k,q,t})/O^D_{i,o,q,t}$ on $\mathcal{C}_{k,t,q}$; positive values favor the direct route.
Settlement objects and operators		
$\mathcal{R}_{k,w}$	Set of route units	Receipt-audited matched route units using vehicle k in UTC week w.
$\mathcal{R}^{\mathrm{transfer}}_{k,w}$	Set of route units	Members whose receipt logs a transfer of vehicle k; $\mathcal{R}^{\mathrm{transfer}}_{k,w} \subseteq \mathcal{R}_{k,w}$.
$\mathcal{R}^3_g,\ \mathcal{R}^4_g$	Sets of route units	Receipt-audited route units in settlement comparison cell g executed on Uniswap V3 and V4, respectively. A matched cell has both sets nonempty.
$\lvert\mathcal{S}\rvert,\ \mathbf{1}_{\{\cdot\}}$	Count / indicator (0/1)	Cardinality of any finite set $\mathcal{S}$ and an indicator equal to one when its subscripted condition is true.
Dynamic operators		
Δ_τ		Change in a daily variable over the τ days ending at t: $\Delta_\tau X_t = X_t - X_{t-\tau}$.

Variable	Formula	Unit	Data column	Definition
Vehicle-use measures				
$\mathrm{VehicleShare}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\mathrm{IVol}_t}$	Fraction (0–1)	`bridge_share`	Fraction of day-t indirect-route USD volume routed through k.
$\mathrm{AllRouteVehicleShare}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\mathrm{Vol}_t}$	Fraction (0–1)	`all_route_bridge_share`	Fraction of day-t all-route USD volume routed indirectly through k.
$\mathrm{VehicleCountShare}_{k,t}$	$\frac{N^I_{k,t}}{N^I_t}$	Fraction (0–1)	`bridge_count_share`	Fraction of day-t indirect route units that use k.
$\mathrm{PairCoverage}_{k,t}$	$\frac{\lvert\mathcal{A}^k_t\rvert}{\lvert\mathcal{A}_t\rvert}$	Fraction (0–1)	`pair_coverage`	Fraction of active indirect-route endpoint pairs that use k.
$\mathrm{MainVehiclePairShare}_{k,t}$	$\frac{\lvert\mathcal{M}^k_t\rvert}{\lvert\mathcal{A}_t\rvert}$	Fraction (0–1)	`pair_main_vehicle_share`	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
$\mathrm{IVol}_{k,t}$		USD	`bridge_volume_usd`	Sum of realized USD volume over indirect routes using vehicle k on day t.
Pair-level route-choice measures				
$\mathrm{VehicleShare}_{i,o,k,t}$	$\frac{\mathrm{IVol}_{i,o,k,t}}{\mathrm{IVol}_{i,o,t}}$	Fraction (0–1)	`actual_vehicle_share`	Candidate k's share of realized indirect-route USD volume for ordered pair (i,o) on day t, defined when $\mathrm{IVol}_{i,o,t} > 0$.
$\mathrm{IndirectRouteShare}_{i,o,t}$	$\frac{\mathrm{IVol}_{i,o,t}}{\mathrm{Vol}_{i,o,t}}$	Fraction (0–1)	`pair_indirect_route_share`	Fraction of realized route-unit USD volume for ordered pair (i,o) executed through an indirect route on day t, defined when $\mathrm{Vol}_{i,o,t} > 0$.
Network and route-denominator controls				
$\mathrm{VolShare}_{k,t}$	$\frac{\mathrm{LegVol}^{\mathrm{in}}_{k,t} + \mathrm{LegVol}^{\mathrm{out}}_{k,t}}{\sum_\ell (\mathrm{LegVol}^{\mathrm{in}}_{\ell,t} + \mathrm{LegVol}^{\mathrm{out}}_{\ell,t})}$	Fraction (0–1)	`vol_share`	$\mathrm{LegVol}^{\mathrm{in}}_{k,t}$ and $\mathrm{LegVol}^{\mathrm{out}}_{k,t}$ are input-side and output-side route-leg USD volumes for k; ℓ indexes every token in the day-t network.
$\mathrm{Betweenness}_{k,t}$	$\frac{N^I_{k,t}}{N_t - N^{\mathrm{in}}_{k,t} - N^{\mathrm{out}}_{k,t}}$	Fraction (0–1)	`betweenness_centrality`	Fraction of day-t intent routes on which k is intermediate, excluding routes where k is the input or output endpoint. Superscripts identify each route role.
$\mathrm{Betweenness}^{\mathrm{vol}}_{k,t}$	$\frac{\mathrm{IVol}_{k,t}}{\mathrm{Vol}_t - \mathrm{Vol}^{\mathrm{in}}_{k,t} - \mathrm{Vol}^{\mathrm{out}}_{k,t}}$	Fraction (0–1)	`volume_weighted_betweenness`	USD-volume analogue of $\mathrm{Betweenness}_{k,t}$ using the same route universe and input/output-endpoint exclusions.
Vol_t	$\mathrm{DVol}_t + \mathrm{IVol}_t$	USD	`daily_all_route_volume_usd`	Total realized USD volume across direct and indirect route units on day t.
IVol_t		USD	`daily_indirect_route_volume_usd`	Total realized USD volume across indirect route units on day t.
$\mathrm{IndirectRouteShare}_t$	$\frac{\mathrm{IVol}_t}{\mathrm{Vol}_t}$	Fraction (0–1)	`indirect_route_share`	Fraction of day-t all-route USD volume executed through indirect routes.
Liquidity measures				
$L_{k,t}$	$\sum_{p \in \mathcal{L}_{k,t}} \frac{\mathrm{TVL}_{p,t}}{m_p}$	USD	`vehicle_linked_liquidity_usd`	Candidate k's allocated share of valid Uniswap V3 pool TVL: full TVL when k is the pool's only candidate token and one half when both pool tokens are candidates.
$\mathrm{LPConc}_{k,t}$	$\frac{L_{k,t}}{\sum_{\ell \in \mathcal{K}} L_{\ell,t}}$	Fraction (0–1)	`lp_concentration`	Candidate k's share of total vehicle-linked liquidity on day t.
$\mathrm{LogVehicleLiquidity}_{k,t}$	$\ln(1 + L_{k,t})$	Natural-log points	`log_vehicle_linked_liquidity`	Natural log of one plus $L_{k,t}$ measured in USD.
Liquidity commonality measures				
$\mathrm{VehicleLiquidityFactor}_{p,k,t}$	$\frac{\sum_{p' \in \mathcal{L}_{k,t} \setminus \{p\}} \Delta_1 \ln(\mathrm{TVL}_{p',t})}{\lvert\mathcal{L}_{k,t} \setminus \{p\}\rvert}$	Daily log-change points	`vehicle_factor_loo`	Leave-one-out mean daily log TVL change among other valid pools linked to candidate k; requires at least three other pools.
$\mathrm{MarketLiquidityFactor}_{p,t}$	$\frac{\sum_{p' \in \mathcal{L}_t \setminus \{p\}} \Delta_1 \ln(\mathrm{TVL}_{p',t})}{\lvert\mathcal{L}_t \setminus \{p\}\rvert}$	Daily log-change points	`market_factor_loo`	Leave-one-out mean daily log TVL change across all other valid pools in $\mathcal{L}_t$; market-wide control for the vehicle factor.
Route-cost opportunity measures				
$\mathrm{AnyIndirectAvailable}_{i,o,q,t}$	$\mathbf{1}_{\{\sum_{k \in \mathcal{K} \setminus \{i,o\}} I_{i,o,k,q,t} > 0\}}$	Indicator (0/1)	`any_indirect_available`	Equals one when at least one candidate in $\mathcal{K} \setminus \{i,o\}$ provides an executable indirect route for pair (i,o) at input notional q.
$\mathrm{DirectDepth}_{i,o,q,t}$	$\frac{O^D_{i,o,q,t}}{q}$	Output USD per input USD	`pair_direct_depth`	Direct-route output ratio for cells with $D_{i,o,q,t} = 1$; undefined when the direct route is not executable.
$\mathrm{IndirectDepth}_{i,o,k,q,t}$	$\frac{O^I_{i,o,k,q,t}}{q}$	Output USD per input USD	`pair_indirect_depth`	Indirect-route output ratio through candidate k for $(i,o) \in \mathcal{I}_{k,t,q}$; undefined when that route is not executable.
$\mathrm{DirectAvailable}_{k,t,q}$	$\frac{\lvert\mathcal{D}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`direct_available_share`	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $D_{i,o,q,t} = 1$. The un-suffixed data column fixes $q = \$10{,}000$.
$\mathrm{IndirectAvailable}_{k,t,q}$	$\frac{\lvert\mathcal{I}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`vehicle_available_share`	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $I_{i,o,k,q,t} = 1$, so both legs through k are executable.
$\mathrm{IndirectOnlyAvailable}_{k,t,q}$	$\frac{\lvert\mathcal{I}_{k,t,q} \setminus \mathcal{D}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`no_direct_vehicle_available_share`	Fraction of pairs with $D_{i,o,q,t} = 0$ and $I_{i,o,k,q,t} = 1$: no executable direct route but an executable route through k.
$\mathrm{DirectDepth}_{k,t,q}$	$\mathrm{median}_{\mathcal{D}_{k,t,q}} \left(O^D_{i,o,q,t}/q\right)$	Output USD per input USD	`direct_depth_median`	Median direct-route output ratio $O^D_{i,o,q,t}/q$ over executable-direct pairs $(i,o) \in \mathcal{D}_{k,t,q}$.
$\mathrm{DirectCostAdvantage}_{k,t,q}$	$\mathrm{median}_{\mathcal{C}_{k,t,q}} \Delta C^D_{i,o,k,q,t}$	Fraction	`direct_cost_advantage_median`	Median $\Delta C^D_{i,o,k,q,t}$ over common-support pairs $(i,o) \in \mathcal{C}_{k,t,q}$; positive values favor the direct route and negative values favor the indirect route through k.
$\mathrm{IndirectBeatsDirect}_{k,t,q}$	$\frac{\lvert\mathcal{W}_{k,t,q}\rvert}{\lvert\mathcal{C}_{k,t,q}\rvert}$	Fraction (0–1)	`vehicle_beats_direct_share`	Fraction of $(i,o) \in \mathcal{C}_{k,t,q}$ for which $\Delta C^D_{i,o,k,q,t} < 0$, equivalently $O^I_{i,o,k,q,t} > O^D_{i,o,q,t}$.
$\mathrm{ThinDirectShare}_{k,t,q}$	$\frac{\lvert\mathcal{T}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$	Fraction (0–1)	`thin_direct_share`	Fraction of pairs with an executable direct quote but $O^D_{i,o,q,t}/q < 0.9$. This is a quote-quality proxy, not a direct measure of pool liquidity depth.
Stress and dynamic variables				
$\mathrm{CandidateStress}_{k,t}$	$\max\left\{-\frac{R_{k,t}}{\sigma^{(30)}_{k,t-1}}, 0\right\}$	Trailing-volatility standard deviations	`candidate_downside_stress`	Candidate-specific adverse return shock scaled by trailing volatility. The measure captures crashes for volatile candidates and downward depegs for stable candidates.
Stress_t	$\max\{-R^{\mathrm{WETH}}_t, 0\}$	Daily log-return fraction	`stress_downside`	Positive part of the negative day-t WETH log return.
$\mathrm{StressEvent}_t$	$\mathbf{1}_{\{\mathrm{Stress}_t \ge 0.08\}}$	Indicator (0/1)	`stress_event_8pct`	Equals one when Stress_t is at least 0.08 on day t.
$\Delta_\tau \mathrm{VehicleShare}_{k,t}$	$\mathrm{VehicleShare}_{k,t} - \mathrm{VehicleShare}_{k,t-\tau}$	Fraction points	`delta_bridge_share_t7`	τ-day change for vehicle k ending on day t. The displayed data column is the $\tau = 7$ instance; the observations table also constructs $\tau \in \{1, 14, 30\}$.
Persistence and displacement measures				
$\mathrm{Incumbent}_{i,o,k,t}$	$\mathbf{1}_{\{k = k^\star_{i,o,t}\}}$	Indicator (0/1)	`incumbent_vehicle`	Equals one when candidate k is the trailing-30-day incumbent for ordered pair (i,o); the ranking window ends at $t-1$.
$\mathrm{ChallengerCostEdge}_{i,o,q,t}$	$\frac{O^I_{i,o,h^\star,q,t} - O^I_{i,o,k^\star,q,t}}{O^I_{i,o,k^\star,q,t}}$	Fraction	`challenger_cost_edge`	Quoted-output advantage of the best executable challenger over the incumbent vehicle at input notional q; positive values favor the challenger.
$\mathrm{VehicleSwitch}_{i,o,q,t,\tau}$	$\mathbf{1}\left\{\begin{array}{c}\mathrm{VehicleShare}_{i,o,h^\star,t+\tau}\\ > \mathrm{VehicleShare}_{i,o,k^\star,t+\tau}\end{array}\right\}$	Indicator (0/1)	`vehicle_switch`	Equals one when the day-t best challenger has a larger pair-level vehicle share than the day-t incumbent at exact calendar horizon $t+\tau$.
Execution-architecture measures				
$\mathrm{DirectConstraint}^{\mathrm{pre}}_{i,o,q}$	$1 - \frac{\sum_{u \in \mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}} D_{i,o,q,u}}{\lvert\mathcal{T}^{\mathrm{V3}}_{\mathrm{pre}}\rvert}$	Fraction (0–1)	`pre_v3_direct_constraint`	Fraction of the fixed 180-day pre-V3 window in which pair (i,o) lacks an executable direct route at notional q; higher values mean a more constrained pre-architecture direct market.
$\mathrm{PostV3}_t$	$\mathbf{1}_{\{t \ge t^{\mathrm{V3}}_0\}}$	Indicator (0/1)	`post_v3`	Equals one on and after the Uniswap V3 activation date.
V4 settlement implementation measures				
$\mathrm{Transfer}_{r,k}$	$\mathbf{1}_{\{r \in \mathcal{R}^{\mathrm{transfer}}_{k,w}\}}$	Indicator (0/1)	`has_matching_transfer`	Equals one when route unit r's transaction receipt contains an ERC-20 Transfer log matching intermediate vehicle k.
$\mathrm{V4}_r$	$\mathbf{1}_{\{r \in \mathcal{R}^4_g\}}$	Indicator (0/1)	`v4_route`	Equals one when matched route unit r executes on Uniswap V4.
$\mathrm{V4RouteShare}_g$	$\frac{\lvert\mathcal{R}^4_g\rvert}{\lvert\mathcal{R}^3_g\rvert + \lvert\mathcal{R}^4_g\rvert}$	Fraction (0–1)	`v4_route_share`	Fraction of receipt-audited route units in comparison cell g executed on V4; reported alongside transfer incidence to distinguish route use from token movement.
$\mathrm{TransferIncidence}_{k,w}$	$\frac{\lvert\mathcal{R}^{\mathrm{transfer}}_{k,w}\rvert}{\lvert\mathcal{R}_{k,w}\rvert}$	Fraction (0–1)	`settlement_transfer_incidence`	Fraction of receipt-audited route units $r \in \mathcal{R}_{k,w}$ containing an ERC-20 Transfer log for intermediate vehicle k.
$\mathrm{ReceiptCount}_{k,w}$	$\lvert\mathcal{R}_{k,w}\rvert$	Route-unit count	`settlement_receipt_count`	Number of receipt-audited matched route units for vehicle k in UTC week w.